



New User Seminar

Website: www.rc.colorado.edu

Documentation: curc.readthedocs.io

Support request form: colorado.service-now.com/req_portal?id=ucb_sc_rc_form



Research Computing
UNIVERSITY OF COLORADO **BOULDER**

Meet the User Support Team



Layla
Freeborn



Brandon
Reyes



Andy
Monaghan



Michael
Schneider



John
Reiland



Dylan
Gottlieb



Mohal
Khandelwal

RC Resources

Tech Support

- High-Performance Computing (HPC)
- Data Management
- Cloud Computing
- Secure Research

Human Support

- Training Materials & Workshops
- Consultations & Office Hours
- Help Tickets

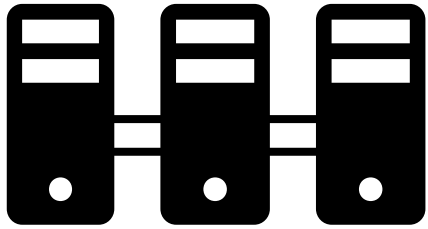


Ask Questions

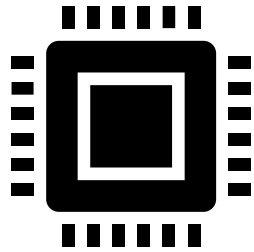


Discuss Ideas

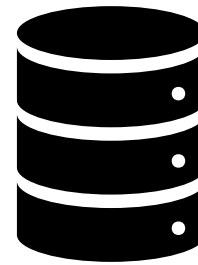
Learning Goals



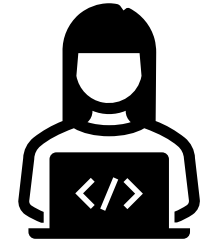
**HPC
Overview**



**Nodes &
Partitions**



**Data
Storage**



**System
Access**

High-Performance Computing



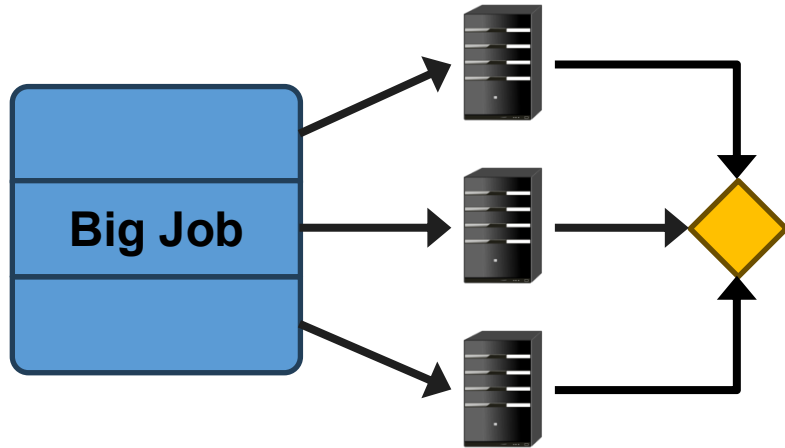
Scale

vs

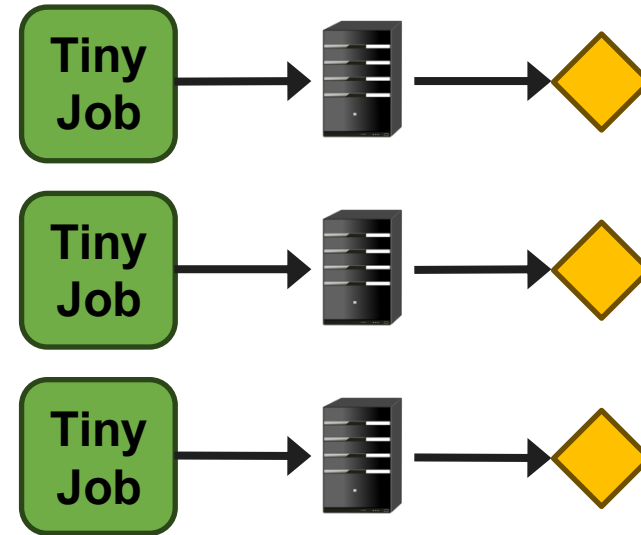
Speed



What can I use HPC for?

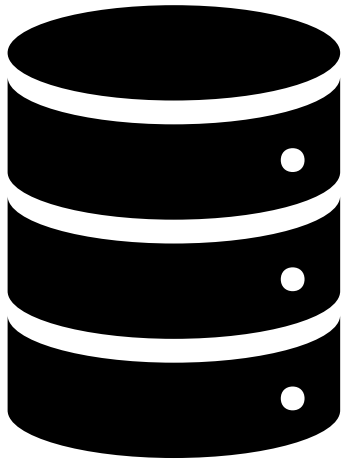


Parallel Jobs

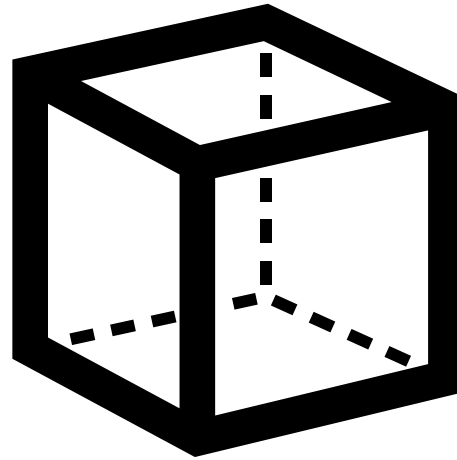


Serial Jobs

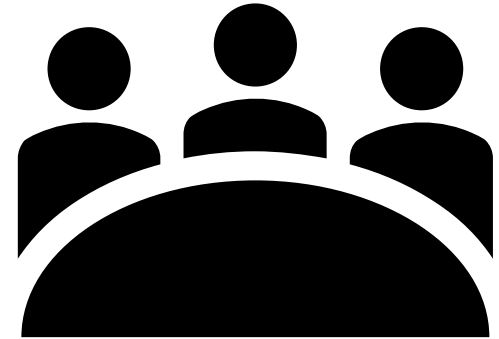
What can I use HPC for? Cont.



Big Data



Viz/Rendering



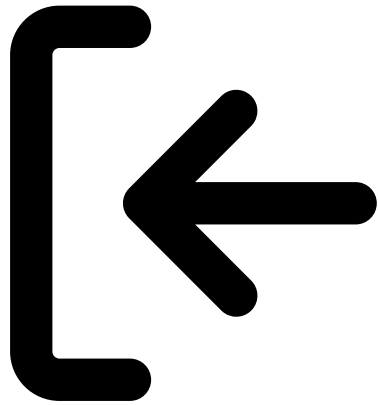
**Shared Work
Environment**

HPC Cluster: Alpine

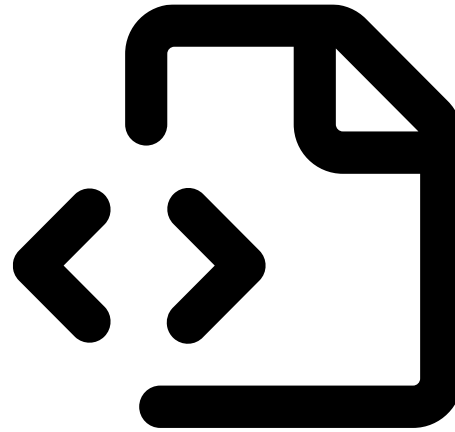


- Heterogeneous cluster
- Hardware provided by CU Boulder, CSU, and AMC
- Access available to CU Boulder, CSU, AMC and RMACC users

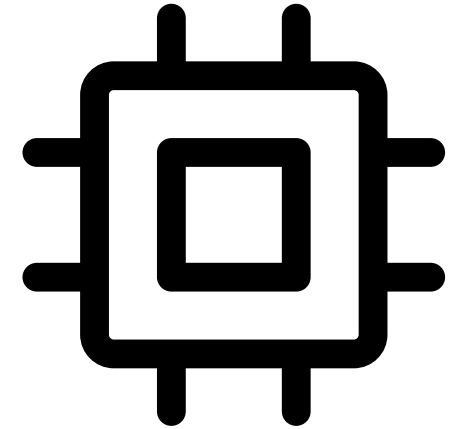
Node Types



LOGIN

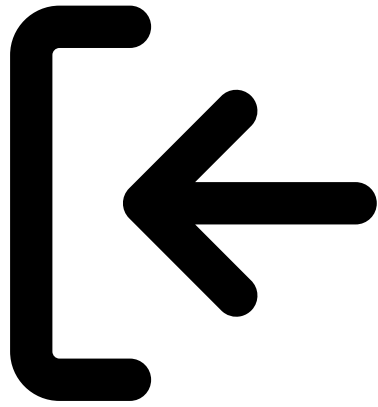


COMPILE



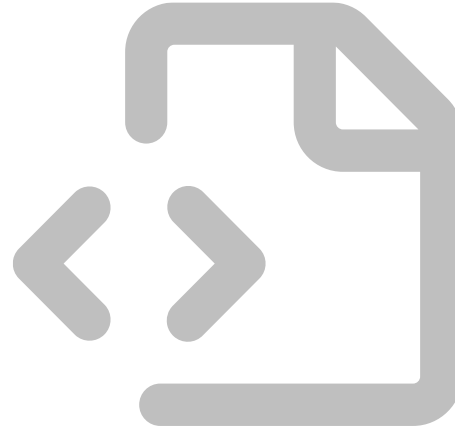
COMPUTE

Node Types (Login Nodes)

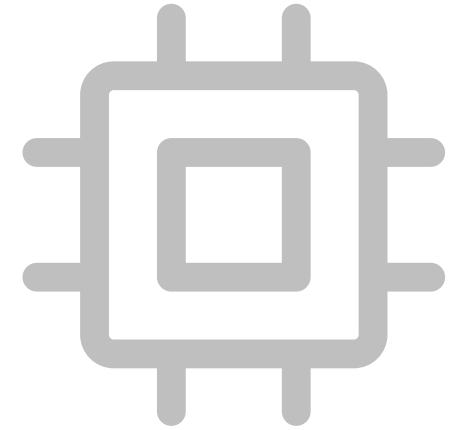


LOGIN

- Entry to system
- View or edit files
- Submit jobs

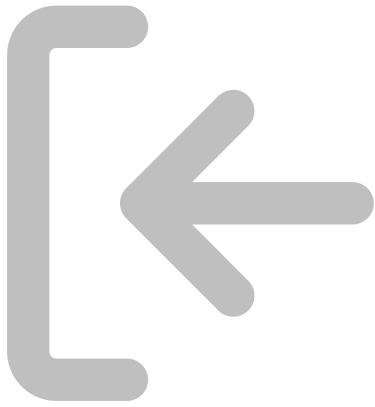


COMPILE

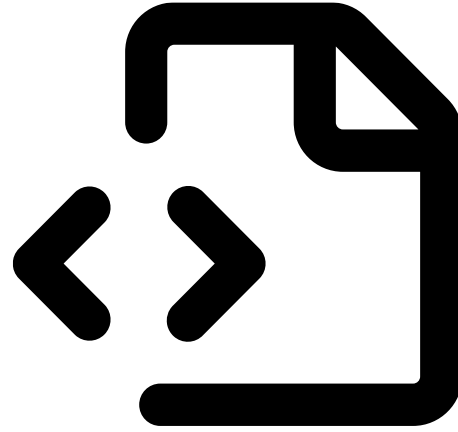


COMPUTE

Node Types (Compile Nodes)

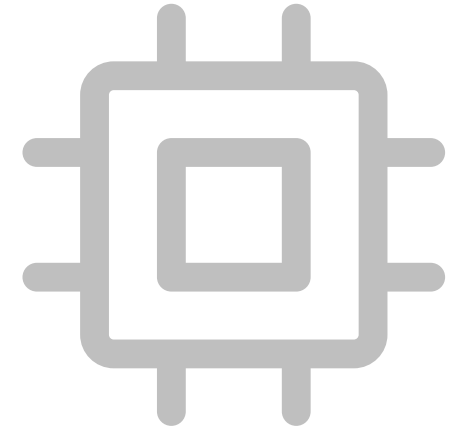


LOGIN



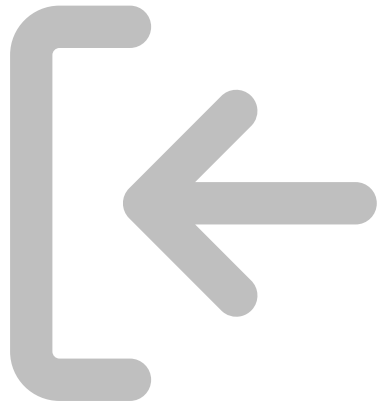
COMPILE

- View or edit files
- Submit jobs
- **Compile code**
- **Install software**

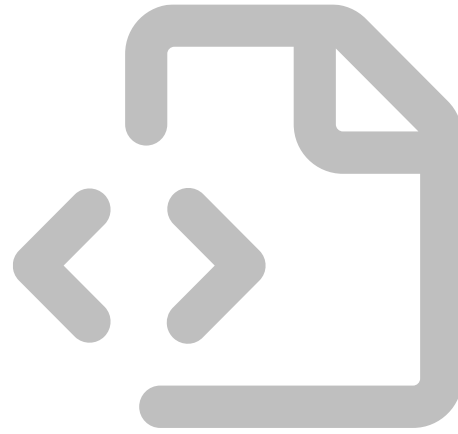


COMPUTE

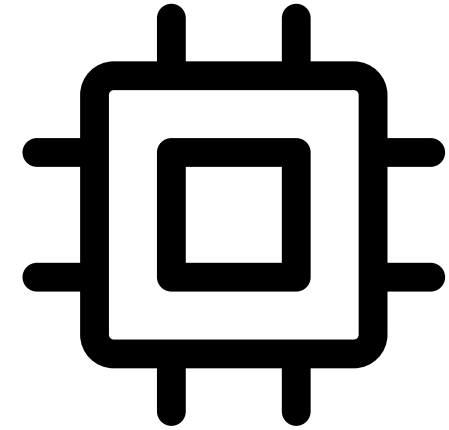
Node Types (Compute Nodes)



LOGIN



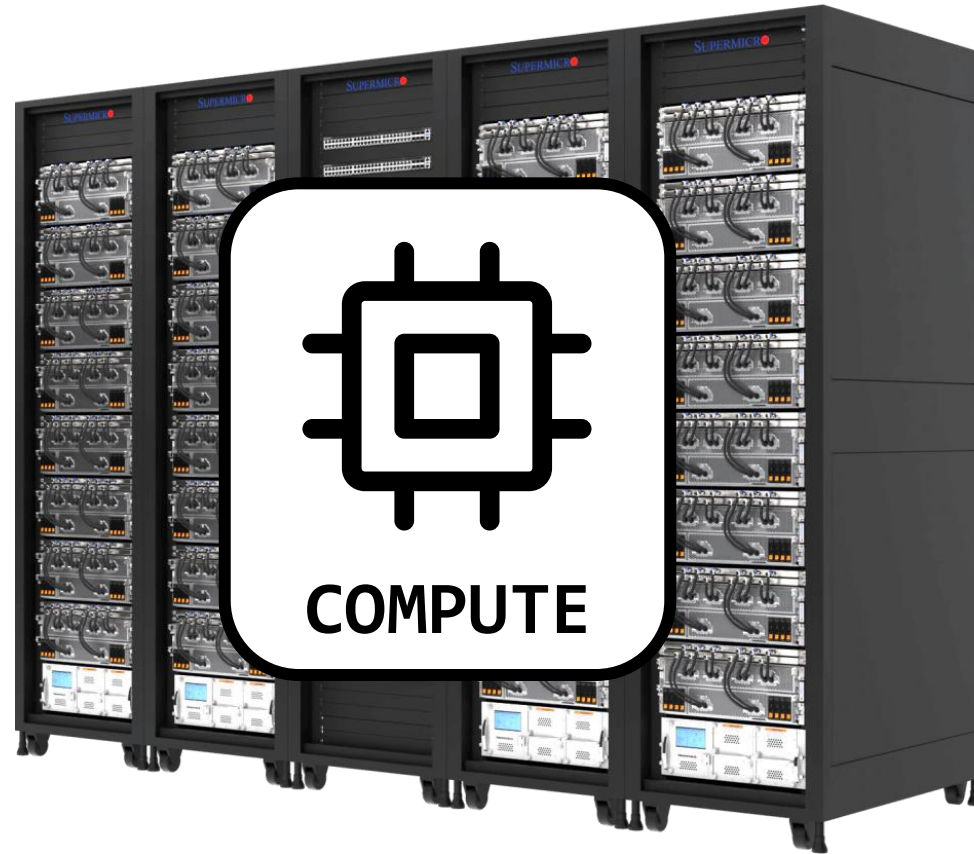
COMPILE



COMPUTE

- Run scheduled jobs
- Handle calculations

Alpine Partitions



Alpine Partitions - acpu

acpu

General Usage



Alpine Partitions - amem

acpu

General Usage

amem

High Memory



Alpine Partitions – NVIDIA GPUs

acpu

General Usage

amem

High Memory



aa100

al40

ah200

artxpro6000

Nvidia GPUs

Alpine Partitions – AMD GPUs

acpu

General Usage

amem

High Memory



aa100

al40

ah200

artxpro6000

Nvidia GPUs

ami100

AMD GPUs

Data Storage

Core

- Personal Storage
- Includes 3 Directories
 - /home (2 GB)
 - /projects (250 GB)
 - /scratch (10 TB)

PL

- PetaLibrary
- Tiered Storage
 - Active, Archive
- Paid storage
- Starts at 1 TB

Core

Home

- Personal data
- Config files
- Do not share!

2 GB

Projects

- Code, Job scripts
- Installed software
- Shared data

250 GB

Scratch

- Job data (in/out)
- Shared data
- Deleted after 90 days

10 TB

PetaLibrary

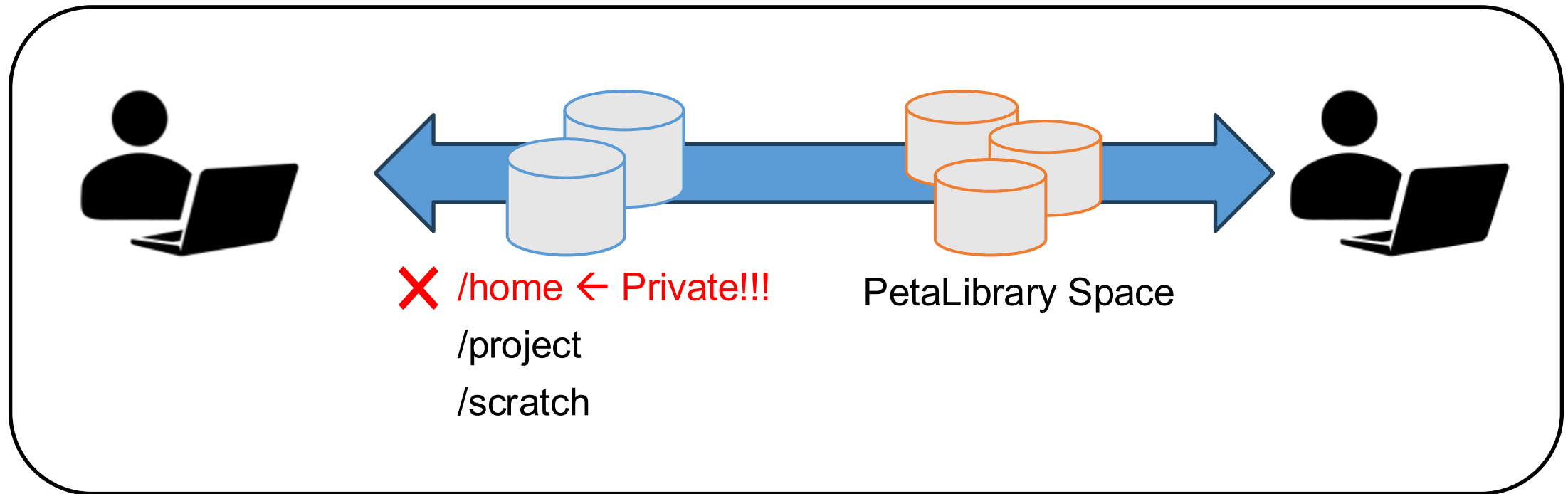
Active

- Performance tier
- Accessible by all nodes
- No file limit

Archive

- Integrity tier
- Accessible by login node
- 10,000 file limit

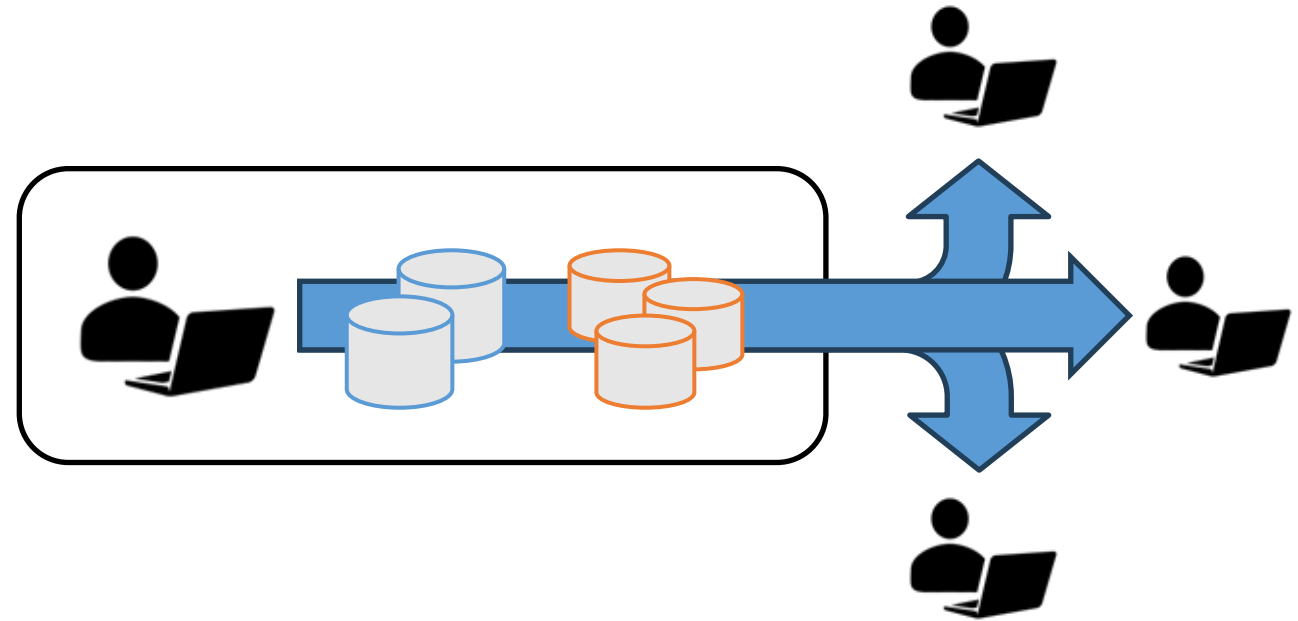
Data Sharing: Within RC



Data Sharing: Outside RC

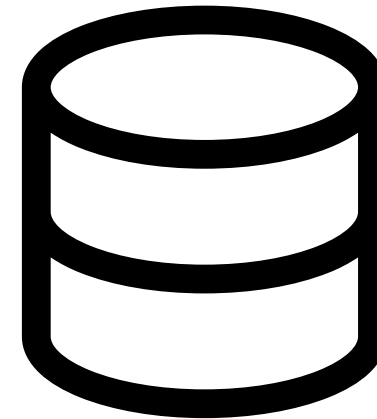
Tools for data transfers:

- Globus (Recommended)
- Open OnDemand (for data <1GB)
- Terminal/Command Line:
 - rsync
 - rclone
 - sftp
 - scp



Data Transfer Nodes (DTN)

- Dedicated nodes that support data transfers
- Can be selected when using scp, sftp, or ssh transfers
- *dtn.rc.colorado.edu*



DTNs

Acceptable data storage and use

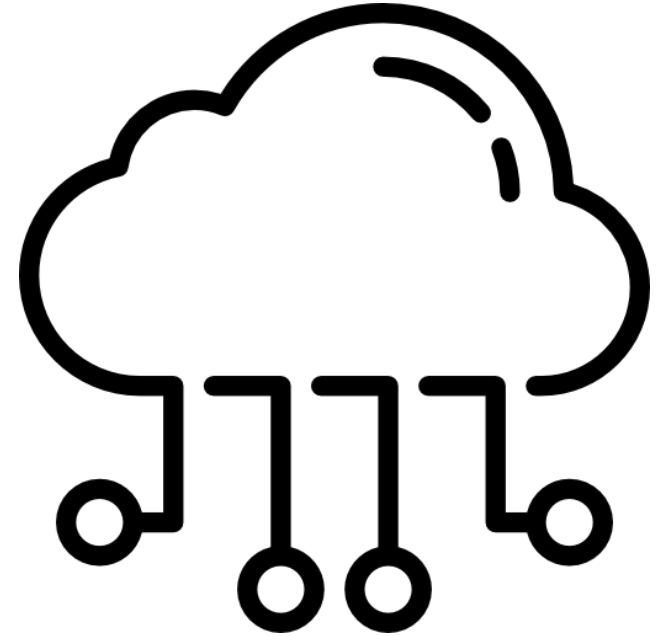
CURC systems and services **should not be used to store** any data that is US government Classified, nor any Controlled Unclassified Information.

For users requiring storage for sensitive data types, please see the secure research computing resources:

www.colorado.edu/rc/secure-research-computing-resources

Cloud Computing

- CURC supports AWS, Azure, and GCP
- Alternative or augmentation to HPC
- Interested and need help? Submit a support request form: colorado.service-now.com/req_portal?id=ucb_sc_rc_form



How to Access RC Resources?

1. Get an RC account
2. Set up two-factor authentication (Duo and/or Microsoft MFA)
3. Log in via 1) *terminal* or 2) *Open OnDemand*
4. Create greatness! (responsibly)

Getting an Account

Visit the appropriate tab under “Getting a CURC account” in the CURC documentation and follow the provided instructions:

curc.readthedocs.io/en/latest/getting_started/logging-in.html#getting-a-curc-account

Getting a CURC account

All individuals wanting to access CU Research Computing (CURC) HPC systems (Alpine and Blanca), Open OnDemand, and storage solutions must have an associated CURC account. CURC systems are utilized by a variety of institutions including CU Boulder, Anschutz Medical Campus (AMC), Colorado State University (CSU), and institutions that are members of the Rocky Mountain Advanced Computing Consortium (RMACC). Given that CURC systems host a variety of institutions, each institution has its own method for creating a CURC account. To simplify this process, below we provide tabs that specify instructions on getting a CURC account based on the institution you are affiliated with.

CU Boulder CSU AMC RMACC

The process for obtaining a CURC account for CU Boulder users is straightforward. Simply navigate to our [Account Request](#) page, select “University of Colorado Boulder”, fill out the provided information, and select “Verify & Continue”. Once the form has been completed and submitted, an automatic process will provision your account. Once you’ve received a CURC account, you will need multi-factor authentication (MFA) to access Research Computing resources. CU Boulder uses two MFA systems depending on how you connect:

- **Web portal services** (such as [Open OnDemand](#) and [Globus](#)) use **Microsoft MFA**. Use the same Microsoft MFA method you already have configured for campus services such as Outlook and Teams. Visit OIT’s [Microsoft 365 Multi-Factor Authentication](#) documentation if you need to enroll.
- **SSH and login-node access** (including [ssh](#), [scp](#), [sftp](#), and connections to data transfer nodes) use **Duo MFA**. After installing the Duo mobile app to your phone (available via Apple App Store or Google Play Store), visit <https://duo.colorado.edu> to enroll.

Important

- Research Computing accounts are separate from your CU accounts. You must sign up for an individual Research Computing account if you wish to access Research Computing resources.
- Once your account is provisioned, please wait 15 minutes before signing in
- If you are having issues with DUO, please consult our [Duo Frequently Asked Questions](#) section
- If you are having issues with Microsoft MFA, contact the [IT Service Center](#) or visit [Buff Techs](#). You will be required to have your Buff OneCard or government ID for MFA help.

Your RC Account

Access to:

1. Alpine Cluster
2. Core Storage
3. PetaLibrary Storage*
4. Open OnDemand
5. Approximately 2,000 Service Units (SUs) per month

**If purchased*

Demo: logging in with Open OnDemand

CURC Open OnDemand is a browser based, integrated, single access point for all of your HPC resources at CU Research Computing.

- CU Boulder: visit <https://ondemand.rc.colorado.edu>.
- All other Institutions: visit <https://ondemand-rmacc.rc.colorado.edu/>

Institution specific instructions:

https://curc.readthedocs.io/en/latest/open_ondemand/index.html#getting-started-with-open-ondemand

The type of two-factor authentication (Duo/Microsoft MFA) needed is institution dependent.

Terminal Access

A terminal interface is accessible via the “terminal” application in Open OnDemand. This application is hosted on our login nodes. A local terminal that connects to CURC resources is available as follows*:

- Mac or Linux
 - Terminal application
- Windows
 - PuTTY
 - Powershell

```
[user0083@login-ci1 ~]$ pwd
/home/user0083
[user0083@login-ci1 ~]$
```

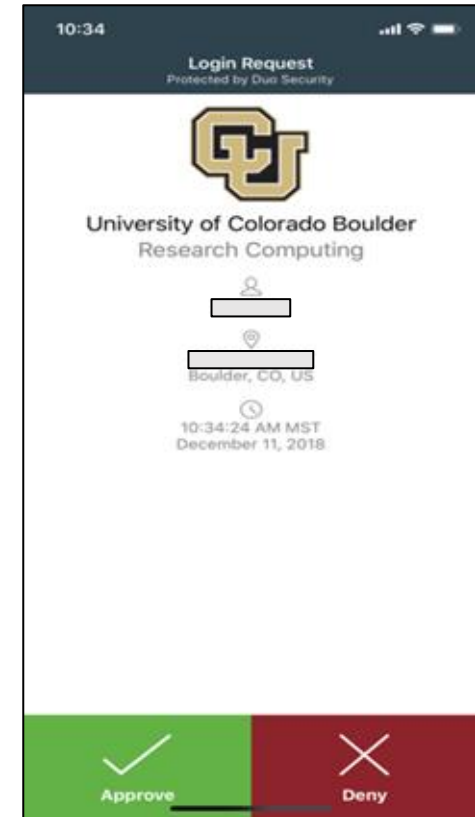
* Local terminal access is exclusive to CU and CSU accounts. Available upon request for AMC. Not available to RMACC.

Duo Authentication

Currently, all ssh access is authenticated through Duo.

Duo authentication can be completed via

- **Duo smartphone app (recommended)**
- Phone Call/Text is an alternatives



Demo: Logging in via Terminal

To login to an RC login node^{*} :

```
$ ssh <username>@login.rc.colorado.edu
```

Supply your institutional password and your Duo app will alert you to confirm the login

^{*} *Exclusive to CU and CSU accounts. Available upon request for AMC. Not available to RMACC.*

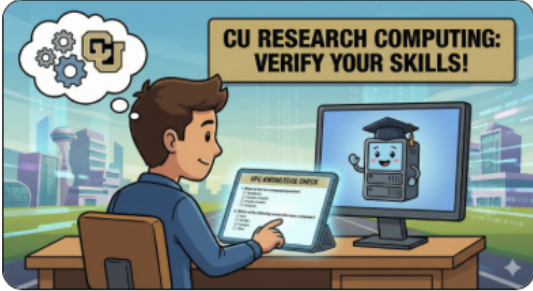
Help! I'm stuck, where do I go?

- **Documentation**: curc.readthedocs.io
- **Trainings with Center for Research Data and Digital Scholarship (CRDDS)**:
www.colorado.edu/crdds
- **Support request form**: colorado.service-now.com/req_portal?id=ucb_sc_rc_form
- **Consult Hours** (Tuesday 1:00-2:00pm in-person or virtually, Thursday 1:00-2:00pm virtually)
- Or....

...try CURC's new self-paced training!

Research Computing New User Training

Welcome to the Research Computing New User Training! This course covers the ins and outs of CU Research Computing's (CURC) High Performance Computing (HPC) systems.



Please enter your Research Computing account information:

Do you have an existing RC account?

Select Yes or No

Institutional email address:

Enter institutional email

Submit

Note: Multiple questions are embedded in this training. It is ok if you don't know the answer to every question! Many of the questions are designed to test for common misconceptions and help you avoid common pitfalls for new users.

[*researchcomputing.github.io/New_User_Async_Training*](https://researchcomputing.github.io/New_User_Async_Training)

Survey and feedback

<https://s.zoom.us/m/bPITUoGhy>

